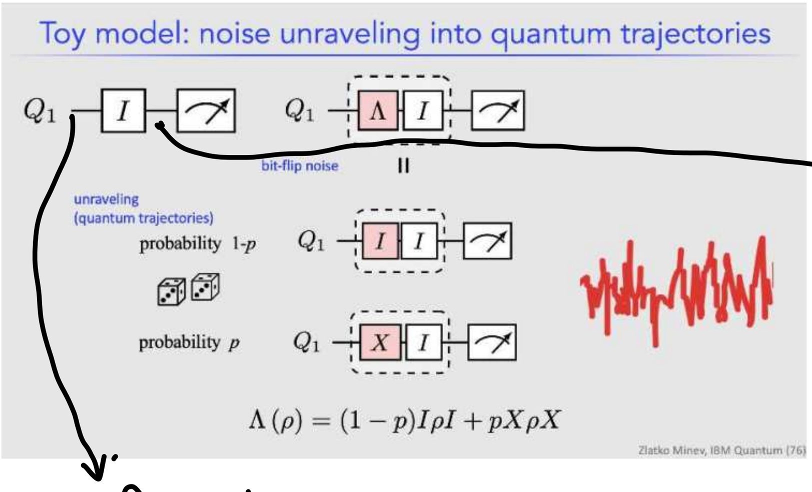
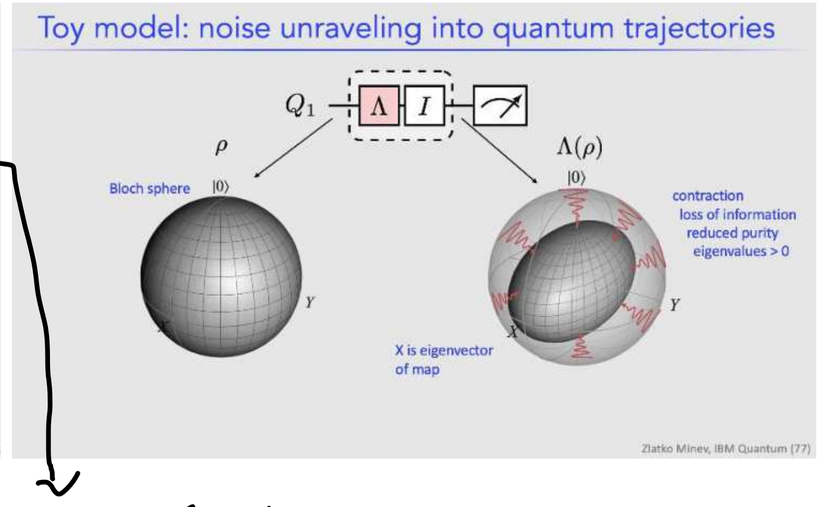
QGSS22 Incoherent noise

Wednesday, June 1, 2022 12:39 PM



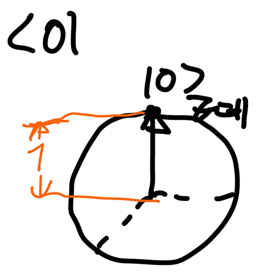


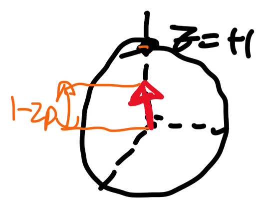
$\rho_0 = |0\rangle\langle 0|$

$\langle X\rangle = 0$

$\langle Y\rangle = 0$

$\langle Z\rangle = 1$





$$\rho = (1-p) I |0\rangle\langle 0| I + p X \rho X$$
$$= (1-p)|0\rangle\langle 0| + p|1\rangle\langle 1|$$
$$= \begin{pmatrix} 1-p & 0 \\ 0 & p \end{pmatrix} = (\cdot\cdot) I + (\cdot\cdot) Z$$

$$\rho_0 = |0\rangle\langle 0| = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = \frac{I+Z}{2}$$

$$\langle X\rangle = \mathrm{Tr}(X\rho_0)$$
$$= \tfrac{1}{2}\mathrm{Tr}(X(I+Z))$$
$$= \tfrac{1}{2}\mathrm{Tr}(X + X\cdot Z)$$

$\mathrm{Tr}(X) = \mathrm{Tr}(Y) = \mathrm{Tr}(Z) = 0$

$\mathrm{Tr}(I) = 2$

$$\langle X\rangle = \tfrac{1}{2}[\cancel{\mathrm{Tr}(X)}^{0} + \cancel{\mathrm{Tr}(X\cdot Z)}^{0}]$$

$X\cdot Z \simeq Y$

$= 0$

$\langle Y\rangle = 0$

$$\langle Z\rangle = \mathrm{Tr}(Z\rho_0)$$
$$= \tfrac{1}{2}\mathrm{Tr}(Z + Z\cdot Z)$$
$$= \tfrac{1}{2}[\cancel{\mathrm{Tr}(Z)}^{0} + \cancel{\mathrm{Tr}(I)}^{2}]$$
$$= 2/2 = 1$$

$\mathrm{Tr}(\rho) = 1$

$$\mathrm{Tr}(\rho^2) = \mathrm{Tr}\begin{pmatrix} (1-p)^2 & \\ & p^2 \end{pmatrix} = (1-p)^2 + p^2$$
$$= 1 - 2p + p^2 + p^2$$
$$= 1 - 2p + 2p^2$$
$$= 1 + 2p(p-1)$$
$$= \begin{cases} 1 & \text{if } p = 0 \\ 1 & \text{if } p = 1 \end{cases}$$

$\langle X\rangle = ? = 0$

$\langle Y\rangle = ? = 0$

$$\langle Z\rangle = \mathrm{Tr}\left(Z\cdot\begin{pmatrix} 1-p & 0 \\ 0 & p \end{pmatrix}\right)$$
$$= \mathrm{Tr}\left(\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\cdot\begin{pmatrix} 1-p & 0 \\ 0 & p \end{pmatrix}\right)$$
$$= \mathrm{Tr}\begin{pmatrix} 1-p & \\ & -p \end{pmatrix}$$
$$= 1 - 2p$$